Order-independent Hermite means: a counterexample and the corrected classification

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Abstract

Conjecture 23 of Horwitz is false: $f(x) = 1/(x+1)$ produces the same mean for every admissible pair of interpolation orders, but that mean is not the harmonic mean. We also classify all generators satisfying the source's nondegeneracy hypotheses at every order: they are shifted reciprocals plus polynomials of degree at most two.

Task 1. The problem and its full resolution

For $0 < a < b$ and integers $0 \le m_2 < m_1$, put $n = m_1 + m_2 + 1$. Let P, Q be the degree-at-most-n Hermite interpolants to f: P matches derivatives through orders m_1, m_2 at a, b, while Q interchanges these orders. Write $m = m_1 - m_2$. The source defines $M_{f,m_1,m_2}(a,b)$ as the unique interior solution of $P = Q$ for even m, and of $2f = P + Q$ for odd m. Equivalently, with $E_P = f - P$ and $E_Q = f - Q$, the equation is

$$E_P(x) = (-1)^m E_Q(x). \tag{1}$$

The existence hypotheses are $f \in C^{n+2}$ and $f^{(n+1)} \neq 0$ on every relevant positive interval. Conjecture 23, on published page 95 of [1], asserts that independence of both orders characterizes $f(x) = C/x$.

Theorem 1. *Suppose $f \in C^\infty(0,\infty)$ and $f^{(r)}(x) \neq 0$ for every $r \ge 3$ and $x > 0$. Its Hermite means are independent of m_1, m_2 if and only if*

$$f(x) = \frac{C}{x+c} + p(x), \qquad C \in \mathbb{R} \setminus \{0\}, \quad c \ge 0, \quad \deg p \le 2. \tag{2}$$

Their common value is

$$M_c(a,b) = \frac{2ab + c(a+b)}{a+b+2c} = \frac{2(a+c)(b+c)}{a+b+2c} - c. \tag{3}$$

In particular, $f(x) = 1/(x+1)$ is a counterexample to Conjecture 23.

Exact proof of sufficiency and of the counterexample

First take $f(x) = C/(x+c)$ and set

$$W_P(x) = (x-a)^{m_1+1}(x-b)^{m_2+1}, \qquad W_Q(x) = (x-a)^{m_2+1}(x-b)^{m_1+1}.$$

The polynomial $C - (x+c)P(x)$ has degree at most $n+1$ and vanishes at the nodes with exactly the required multiplicities, or higher ones. It is therefore a constant multiple of the monic polynomial W_P. Evaluation at $x = -c$ determines that constant and gives

$$E_P(x) = \frac{CW_P(x)}{(x+c)W_P(-c)}, \qquad E_Q(x) = \frac{CW_Q(x)}{(x+c)W_Q(-c)}. \tag{4}$$

The denominators are nonzero since $a, b > 0$ and $c \geq 0$. For $a < x < b$, their ratio is

$$\frac{E_P(x)}{E_Q(x)} = (-1)^m \left(\frac{(x-a)(b+c)}{(b-x)(a+c)} \right)^m .$$

The quantity in parentheses is positive. Thus (1), for either parity and every $m \geq 1$, is equivalent to

$$\frac{x-a}{b-x} = \frac{a+c}{b+c}.$$

This has exactly the solution (3) in (a, b). Every allowed degree n is at least two, so adding a fixed quadratic p adds p to both interpolants and leaves both errors unchanged. This proves sufficiency. The family has $f^{(r)}(x) = (-1)^r r! C/(x+c)^{r+1} \neq 0$ for $r \geq 3$.

For $c = 1, C = 1, p = 0$, the function is not C_0/x for any constant C_0, since $x/(x+1)$ is not constant. Its mean differs from the harmonic mean H and the arithmetic mean A whenever $a \neq b$:

$$M_c - H = \frac{c(a-b)^2}{(a+b)(a+b+2c)}, \qquad A - M_c = \frac{(a-b)^2}{2(a+b+2c)}. \tag{5}$$

For example, $M_1(1,2) = 7/5$, whereas $H(1,2) = 4/3$. Thus the counterexample changes the common mean itself. The source already notes invariance under polynomial additions in Remark 5; that invariance alone is not the basis of this counterexample.

A local coefficient of an arbitrary Hermite mean

For confluent divided differences, let $a^{[j]}$ denote j copies of the node a. The Hermite error formula turns (1) into

$$(u-a)^m D_P = (b-u)^m D_Q, \quad D_P = f[u, a^{[m_1+1]}, b^{[m_2+1]}], \quad D_Q = f[u, a^{[m_2+1]}, b^{[m_1+1]}]. \tag{6}$$

This is equation (4) of the source and is valid for both parities.

Lemma 1. *Fix an allowed pair and $t > 0$. For smooth f with $f^{(n+1)}(t) \neq 0$,*

$$M_{f,m_1,m_2}(t-h, t+h) = t + \frac{f^{(n+2)}(t)}{(n+2) f^{(n+1)}(t)} h^2 + O(h^3) \qquad (h \downarrow 0).$$

Proof. The divided difference of order N at nodes $t+\delta_0, \ldots, t+\delta_N$ has the expansion

$$\frac{f^{(N)}(t)}{N!} + \frac{f^{(N+1)}(t)}{(N+1)!} \sum_{i=0}^{N} \delta_i + O(h^2) \quad \text{if } |\delta_i| \leq h. \tag{7}$$

For completeness, the Hermite–Genocchi integral writes this divided difference as $\mathbb{E} f^{(N)}(t + \sum_i V_i \delta_i)/N!$, where V is uniform on the simplex $\sum_i V_i = 1$, $V_i \geq 0$. Taylor expansion and $\mathbb{E} V_i = 1/(N+1)$ give (7), including repeated nodes, with uniform remainder. The integral itself follows by repeated use of the fundamental theorem of calculus.

Write $u = t + s$, so $|s| < h$, and put $B = f^{(n+1)}(t)/(n+1)! \neq 0$ and $D = f^{(n+2)}(t)/(n+2)!$. The sums of offsets in the two divided differences are $s - mh$ and $s + mh$. Consequently

$$D_P = B + D(s - mh) + O(h^2), \qquad D_Q = B + D(s + mh) + O(h^2).$$

For small h, their ratio is positive and equals $1 + O(h)$. Equation (6) first implies $m \log((h+s)/(h-s)) = O(h)$, hence $s = O(h^2)$. Expanding once more gives

$$2m \frac{s}{h} + O(h^3) = \log \frac{D_Q}{D_P} = 2mh \frac{D}{B} + O(h^2).$$

Multiplication by $h/(2m)$ proves the formula. □

Necessity and the global domain

If the means are independent of the orders, compare $(m_1, m_2) = (1, 0)$ and $(2, 0)$, whose degrees are $n = 2$ and $n = 3$. Lemma 1 implies, at every $t > 0$,

$$\frac{f^{(4)}}{4f^{(3)}} = \frac{f^{(5)}}{5f^{(4)}}.$$

Set $v = f^{(4)}/(4f^{(3)})$, which is smooth and never zero. Differentiating and using the displayed identity gives

$$v' = \frac{f^{(5)}}{4f^{(3)}} - \frac{(f^{(4)})^2}{4(f^{(3)})^2} = 5v^2 - 4v^2 = v^2.$$

Thus $(-1/v)' = 1$, so $v(t) = -1/(t + c)$ for a constant c. The denominator cannot vanish anywhere on $(0, \infty)$; hence $c \geq 0$. The equation $f^{(4)}/f^{(3)} = -4/(t + c)$ now yields $f^{(3)} = K/(t + c)^4$ with $K \neq 0$. Integrating three times gives (2), with $C = -K/6$. This proves the theorem.

The nondegeneracy in the classification is precisely what results from requiring the source's hypotheses for every allowed pair: $n + 1$ runs through all integers at least three. The counterexample satisfies these hypotheses on the entire positive half-line. Homogeneity is defined as an additional property in the source, not a hypothesis of Conjecture 23. If homogeneity is imposed additionally, (3) forces $c = 0$.

Follow-up tasks

These tasks follow the full resolution of Task 1 and are not additional completed or independently screened open problems.

Task 2. Classify order-independent generators on arbitrary real intervals, including intervals on either side of a pole and the appropriate conditions at finite endpoints.

Task 3. Determine which two prescribed order pairs already force a shifted reciprocal plus a polynomial, and identify exceptional pairs.

Task 4. Prove quantitative stability estimates: bound a generator's distance from the classified family using a uniform bound on the difference between its means for two consecutive total degrees.

Task 5. Classify generators whose means become independent of the orders only after a fixed total degree, and determine the resulting polynomial freedom and possible limiting means.

Status and verification

The dated accompanying files record the live open status, the search for earlier public solution claims, and exact checks of confluent interpolation, both parity equations, and the local coefficient. The proof is analytic; the finite checks are supplementary. No independent peer review or proof-assistant certification is claimed.

References

[1] A. Horwitz, *Means and Hermite Interpolation*, Journal of Mathematical Inequalities **2**(1) (2008), 75–95, Conjecture 23. https://doi.org/10.7153/jmi-02-08. https://files.ele-math.com/articles/jmi-02-08.pdf. Preprint: https://arxiv.org/abs/0711.4940.